Project Design Phase-II
Technology Stack (Architecture & Stack)

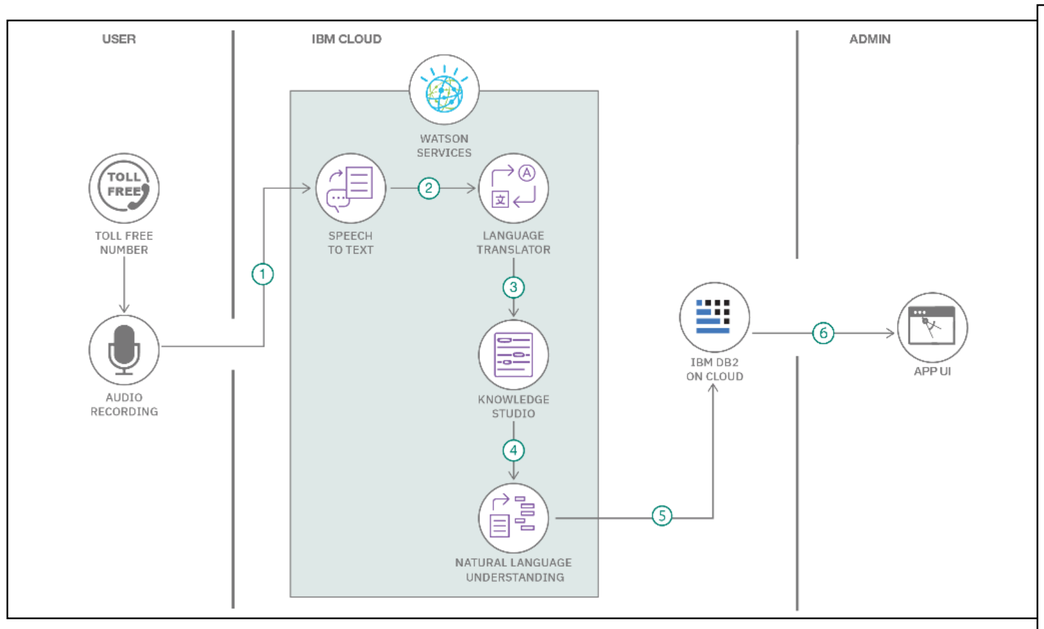
Date	31 January 3035
Team ID	
Project Name	AI Fit Tracker-API
Maximum Marks	4 Marks

Technical Architecture:

The Deliverable shall include the architectural diagram as below and the information as per the table1 & table 2

Example: AI-powered fitness tracking and health activity management

Reference: API-based fitness tracking and AI analysis



Guidelines:

Include all the processes (As an application logic / Technology Block)
Provide infrastructural demarcation (Local / Cloud)
Indicate external interfaces (third party API's etc.)
Indicate Data Storage components / services
Indicate interface to machine learning models (if applicable)

Table-1 : Components & Technologies:

S.No	Component	Description	Technology
User Interface	User interacts with the AI Fit Tracker application through a web/mobile interface.	HTML, CSS, JavaScript / React.js	HTML, CSS, JavaScript / Angular Js / React Js etc.
Application Logic-1	Handles user fitness data, workout records, goals and API requests.	Python / FastAPI	Java / Python
Application Logic-2	Provides fitness and activity analysis through API-based processing.	Python / FastAPI REST API	IBM Watson STT service
Application Logic-3	Generates personalized fitness insights and recommendations.	AI/ML service / Python	IBM Watson Assistant
Database	Stores user profiles, fitness records, workout details and activity history.	MySQL / MongoDB	MySQL, NoSQL, etc.
Cloud Databas	Cloud-hosted database for application data and API access.	MongoDB Atlas / Cloud Database	IBM DB2, IBM Cloudant etc.

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File Storage	Stores profile images or other application files when required.	Cloud Object Storage / Local Filesystem	IBM Block Storage or Other Storage Service or Local Filesystem
External API–1	Provides fitness-related external data or activity information.	Fitness / Health API	IBM Weather API, etc.
External API–2	Provides supporting external services required by the application.	Authentication / Maps API, if applicable	Aadhar API, etc.
Machine Learning Model	Analyses fitness data and supports personalized recommendations.	AI/ML Recommendation Model	Object Recognition Model, etc.
Infrastructure (Server / Cloud)	Deploys the API and application services for user access.	Local Server / Cloud Server	Local, Cloud Foundry, Kubernetes, etc.

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
Open-Source Frameworks	List of open-source frameworks used for the application.	FastAPI, Python, React.js	Technology of Opensource framework
Security	Authentication, authorization, secure API access and input validation.	HTTPS, JWT, password hashing, API validation	e.g. SHA-256, Encryptions, IAM Controls, OWASP etc.
Scalable Architecture	Separates user interface, API/application logic and database services.	3-Tier Architecture, REST API	Technology used
Availability	Application services can be deployed on reliable cloud infrastructure for continuous access.	Cloud hosting, API server	Technology used

Performance	REST APIs, efficient database queries and lightweight data exchange improve response time.	REST API, database indexing, caching where required	Technology used

References:

https://c4model.com/

https://developer.ibm.com/patterns/online-order-processing-system-during-pandemic/

https://www.ibm.com/cloud/architecture

https://aws.amazon.com/architecture

https://medium.com/the-internal-startup/how-to-draw-useful-technical-architecture-diagrams-2d20c9fda90d